Assignment 1

Compiled By: G.P.Lekhak

1. Differentiate between Von Neumann and Harvard architecture.

2. Mention the application of microprocessor.

3. What do you mean by addressing mode? Describe the different types of addressing modes in 8085 microprocessor with suitable at least two example.

4. What is the purpose of flags in microprocessor? Describe different types of flags in 8085.

5. Compare the architecture of 8085 and 8086 microprocessor.

6. Why do we need pipelining? Describe in shorts.

Assignment-2

By G.P Lekhak

1. Draw the timing diagram of MOV A,B , MVI B,45H, IN 01H, OUT40H, LDA 4000H, STA 3000H instructions.
2. Define instruction cycle, machine cycle, and T-states.
3. Describe the instructions according to size with examples.
4. Classify the instructions according to following categories: Arithmetic operation, Logical operation, Data transfer operation, Branching operation, Control operation

Assignment-3

1. WAP to load all register with your class roll number immediately.
2) WAP to add 20 bytes of data stored in memory with starting address of 5050H.
3) WAP to add 10 bytes of data and store the 16-bit result at the end of memory address
4) WAP to transfer the 20 bytes o f data stored in memory having starting address of 2012H to the next memory having starting address of 8000H in reverse order.
5) WAP to multiply the 10H and 14H and show the 16 bit result through any port.
6) WAP to count the positive and negative numbers among 20 bytes of data.
7) There are two tables having 10 data each, WAP to add corresponding numbers and show the result through 00H port.
8) There are 10 numbers in the memory, WAP to add only positive numbers and show the result at outport ports 80H.
9) WAP to count the even or odd numbers among 10 bytes of data stored in memory.
10) WAP to count the no. of 1 present in a byte, assume any byte.
11) WAP to find the largest number among 50 bytes of data stored in memory.
Q) WAP to find the smallest number among 50 bytes of data stored in memory.
12) Explain the 8085 instructions according to the size with examples.
13) Data is stored from 4050H to 405AH . Insert 5 data after 4055 taking from 4040H , but do not lose the previous content.
14) Transfer ten bytes data from 5050H to 5060H only if data is between 30H and 70H else store 00H in the next table.
15) Explain the given instructions with example: LHLD , SHLD, XCHG, RLC,RRC,RAL,RAR.
16) There are address of C000H to C009H where data are stored . WAP in 8085A how many times the data 07H is repeated in the given array of data and store the result at address C004H.
17) WAP to display data between 50H to 80H.
18) What is the value of accumulator and carry flag when following instructions are executed.

MVI A, C5H	MVI A, A7H
ORA A	ORA A
RAL	RAR
RRC	RAL

19) What is the value of register and flag as the following instructions are being executed.

```
MVI A, 80H
ORA A
RAR
```

20) What will be the content of C registers show each step clearly.

```
MVI A, 08H
MVI B, 07H
ADD B

ANI 40H
ORI FEH
ADI 01H
INR A
MOV C, A
HLT
```

```
        MVI A ,7FH
        MVI B, 08H
        MVI D,00H
BACK:   RLC
        JNC NEXT
        INR D
NEXT:   DCR B
        JNZ BACK
        MOV C, D
        HLT
```

Assignment-4

BY: G.P.Lekhak

1) Specify the contents of the registers and the flag status as the following instructions are executed.

```
MVI A, 00H
MVI B, F8H
MOV C,A
MOV D,B
HLT
```

2) Write instructions to load the hexadecimal number 65H in register C, and 92H in the accumulator A. Display the number 65H at port0 and 92H at port1.

3) Write instructions to read the data at input PORT 07H and at PORT 08H. Display the input data from PORT 07H at output PORT 00H, and store the input data from PORT 08H in register B.

4) Specify the output at PORT1 if the following program is executed.

```
MVI B, 82H
MOV A,B
MOV C,A
MVI D,37H
OUT PORT1
HLT
```

4) Specify the contents of the registers and the flag status as the following instructions are executed. Specify also the output at PORT0.

A	B	S	Z	CY	
00	**FF**	**0**	**1**	**0**	initial contents

```
MVI A, F2H
MVI B, 7AH
ADD B
OUT PORT0
HLT
```

5) Specify the register contents and the flag status as the following instructions are executed.

A	C	S	Z	CY	
XX	**FF**	**0**	**1**	**0**	initial contents

```
MVI A, 5EH
ADI A, A2H
MOV C,A
HLT
```

6) Write a program using the ADI instruction to add the two hexadecimal numbers 3AH and 48H and to display the answer at an output port.

7) What is the content of register A and the status of the S and CY flags.

```
MVI A, F8H
SUI 69H.
```

8) Specify the register contents and the flag status as the following instructions are executed.

A	B	S	Z	CY	
XX	**XX**	**X**	**X**	**X**	initial contents

```
SUI A
MOV B,A
DCR B
INR B
SUI 01H
HLT
```

9) Write a program to: a) clear the accumulator
b) add 47H (use ADI instruction)
c) subtract 92H
d) add 64H
e) display the result after subtracting 92H and after adding 64H.

10)
```
MVI A,A9H
MVI B,57H
ADD B
ORA A
```
specify the register contents and the flags status(S, CY, Z).

11) Load the data byte A8H in register C. Mask the high-order bits (D7-D4), and display the low-order bits (D3-D0) at an output port.

12)
```
        MVI A,8FH
        ADI,72H
        JC DISPLAY
        OUT PORT1
        HLT
DISPLAY: XRA A
        OUT PORT1
        HLT
```

PORT1=?

13) Replace the instruction **ADI 72H** by the instruction **SUI 67H** and specify the output in question no (12).

13).
```
        MVI A, BYTE1
        MOV B,A
        SUI 50H
        JC DELETE
        MOV A,B
        SUI 80H
        JC DSPLAY
DELETE: XRA A
        OUT PORT1
        HLT
DISPLAY MOV A,B
        OUT PORT2
        HLT
```

WHAT WILL BE THE RANGE OF DATA DISPLAYED AT PORT2.

14) Explain the following programs:

```
            MVI A,BYTE1
            ORA A
            JM OUTPRT
            OUT 01H
            HLT
OUTPRT:     CMA
            ADI 01H
            OUT 01H
            HLT
```